

Developing a Simplified Early Warning System for Predicting Graduation Outcomes in California Public Schools

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ABSTRACT

The overall California graduation rate does not reveal the same level of disparity in other groups, including students who are English learners, have a disability, or are foster youth. We created an early warning system model to identify high schools that may be experiencing low graduation rates using only school-level publicly available data that is compliant with the Family Educational Rights and Privacy Act. The four California Department of Education 2021-2022 datasets used included Adjusted Cohort Graduation Rate (ACGR), chronic absenteeism, Free or Reduced-Price Meals (FRPM) eligible, and school staffing, all of which were aligned with the Attendance, Behavior, and Course Performance (ABC) framework. High schools identified by the California Department of Education as having graduation rates below 90% were labeled "at-risk," and seven supervised learning algorithms were tested for performance on precision, recall, F1, and PR-AUC. The Random Forest model was found to have the highest performance among the seven models tested (PR-AUC 0.78), and chronic absenteeism, Free or Reduced-Price Meals eligibility, unexcused absences, and still enrolled rates were determined to be the most important variables for predicting graduation rates. These findings indicate that using school-level open data can provide reliable approximations of early warning system

student-level indicators and provide evidence for developing equity-focused tools for identifying schools where graduation rates are less than expected.

KEYWORDS

Early warning system (EWS), graduation rates, chronic absenteeism, FRPM Eligibility, Random Forest, educational equity, socioeconomic disadvantage, California Department of Education (CDE)

1 Introduction

Earning a high school diploma is an important milestone for students, as it gives them significant opportunities and helps with a smoother and more successful transition into adulthood, especially for those who might otherwise have a difficult time. Despite its importance, there are thousands of students each year who do not earn a high school diploma in California, which limits their access to higher education and long-term opportunities. The California graduation rate for the 2021-2022 academic year was 87% (California Department of Education, 2023). However, there are subgroups of students who remain under the 80% for graduation rates, including English learners (76.2%), students with disabilities (75.3%), and foster youth (66.7%). These gaps show the need for data-driven tools

that can help schools understand where graduation rates are falling behind and provide the required support.

Many public schools are using early warning systems (EWS) to help identify students at risk of dropping out by tracking attendance, behavior, and course performance. However, these systems often rely on student-level data, which are protected under the Family Educational Rights and Privacy Act (FERPA), limiting the ability for districts and researchers to produce interventions that are at a larger scale (U.S. Department of Education, 2023). To address these challenges, this study proposes the development of a simplified EWS that predicts the graduation outcomes for California public high schools. The system uses school-level datasets that are publicly available from the California Department of Education (CDE) and applies machine learning techniques to identify schools that are at risk of low graduation rates.

The hypothesis is that schools with a graduation rate of less than 80% have a higher rate of chronic absenteeism, a larger percentage of students from low-income families, and a lower average amount of teaching experience among school employees. The goal is to help schools and education leaders have a better understanding of what factors have an influence on graduation rates and where additional support is most needed.

2 Background

Attaining a high school diploma has significant positive impacts on students and their lives after high school (Krueger et al., 2015). Without a diploma, students face more barriers than graduates, including fewer opportunities and a lack of qualification, creating challenges in achieving success. Although this is well known, there is still an abundance of students who drop out, making them more vulnerable to unemployment and lower lifetime earnings.

In California, the graduation rate for the academic year of 2021-2022 was 87%

(California Department of Education, 2023). The California Department of Education measures graduation outcomes using the 4-year adjusted cohort graduation rate (ACGR). The ACGR shows the percentage of students who receive their high school diploma within 4 years of entering the ninth grade and is adjusted for those who transfer in later or transfer out (California Department of Education, 2023). When looking at students who have a disadvantage, the graduation rate drops to less than 80%. This has been consistent throughout the years, especially among the same groups of students.

The EWS is recommended and implemented in many California public schools. This system was developed to monitor indicators that can signal student disengagement, absenteeism, and eventual school dropout (O'Cummings & Therriault, 2015). To foster student retention and academic success, the EWS will examine attendance patterns, behavioral issues, and course performance (the ABC model) and promote intervention when these risk factors are identified (Rumberger et al., 2017). Although the EWS was developed in part with dropout prevention as one of its goals and is valid in predicting dropouts, it is difficult to scale such an effort because of the difficulty of complying with federal and state data privacy regulations.

These consistent outcomes in California present researchers with the opportunity to create data-informed strategies that are accessible and give the necessary information for schools to better address and understand the graduation disparities across California.

2.1 Problem Identification and Motivation

California continues to see graduation outcomes that are imbalanced among subgroups of students, including English learners, foster youth, and students with disabilities. Each academic year, the state tracks this data with CDE and reports how the

schools are performing through the California School Dashboard. These tools provide information on graduation rates, absenteeism, and academic progress, which allows schools to track their overall performance and adjust accordingly for the future school years. However, it does not provide insight into risk factors that are currently affecting students or predict future outcomes.

Many California public secondary schools experience a large gap in their graduation rates, but have no easily accessible, data-based methods to identify the systemic causes of this disparity in graduation rates. Although EWS are being used by many California Public schools, they rely on student-level data that is restricted from the general public. The reason for this restriction is due to FERPA, a federal privacy law that protects individual student records. Due to this, it makes it difficult for districts and researchers to access the detailed data required to build traditional EWSs. This research aims to develop a school-level predictive model with only the publicly available CDE datasets, allowing schools to be identified as at-risk of having a graduation rate below 80%. The proposed approach has the potential to provide schools with the time and insight to intervene before the graduation rates decline.

The ABC model will be applied to track early warning signs of potential students at risk of dropping out of school using the publicly available CDE data for the 2021-22 academic year. ACGR, absenteeism, California basic educational data system, and student poverty measures will be used to provide a statewide and replicable means of identifying early warning patterns of potential dropouts from schools.

The goal is to create an accessible, data-driven EWS for California's education leaders so they can monitor and provide support to their schools through publicly available open data systems (data that does not include personally identifiable information about students) rather than through private student records. This system will align with California's equity and

accountability goals as it utilizes publicly accessible data in a standardized format.

2.2 Definition of Objectives

To address the challenges described above, this section defines the specific objectives and key actions that will guide the development of the proposed EWS. The project will use multiple CDE school-level datasets, which are all publicly accessible. The key actions to achieve this include identifying publicly available data sources, developing a school-level classification model, analyzing school-level predictors of graduation risk, and creating visuals that provide insight to support decision-making.

The objectives of this project focus on using publicly available CDE datasets to build an effective school-level EWS. First, the project identifies and compiles datasets containing key school-level indicators related to graduation outcomes. Second, it develops a school-level classification model designed to predict whether a school is at risk of having a graduation rate below 80%, providing an alternative to a traditional student-level EWS. Third, the project examines school conditions and indicators to determine which factors negatively affect graduation rates and contribute to broader disparities. Finally, the project produces visual insights that highlight high-risk schools and key drivers of dropout rates. With these insights, educators and districts will be able to guide earlier and more targeted interventions.

Through these actions, the project aims to determine meaningful and actionable insights on graduation disparities that can be achieved by publicly available school-level data. The work seeks to identify which school conditions have the most effect on graduation disparities and provide insights that support early and more targeted interventions before graduation rates start to drop. These findings will also have the potential to demonstrate how open, non-sensitive data can be used for equitable decision-making and can uphold accountability efforts across the California education system. If all objectives are not fully met, the study will

still provide valuable insights into the limitations concerning the use of public datasets to predict statewide risk.

3 Literature Review

Educational EWS, the predictive analytical systems used for identifying individual students at risk of not completing high school studies, has recently become a very interesting topic of study in education. Research shows that the ABC model, tracking attendance, behavior, and performance, is a good predictor of future graduation results (O'Cummings & Therriault, 2015; Rumberger et al., 2017).

Nevertheless, all of the currently existing EWS models are based exclusively on student-level data, which is subject to protection under FERPA regulation and thus cannot be made public. Due to the lack of transparent information, the scalability and the possibility of external verification of the EWS models currently being developed are limited, because researchers and districts cannot make the data on which these models are based open to examination or reproduction. Therefore, there is presently no evidence that school-level data can approximate the predictive potential of student-level data, creating a significant void in the literature.

This study will directly address this void by assessing whether publicly available indicators at the school level can provide effective early warning predictions. Below are the major research informing this inquiry, such as Chen's (2019) research on parental involvement, the What Works Clearing House (WWC, 2017) research on effective dropout prevention practices, and Cobb's (2020) research on geographic trends in educational equity.

3.1 Parental Engagement in K–12 Online Learning

Parental participation is a key component for a student's success, regardless of whether they are in an online or technology-based

environment, according to Chen (2019). Chen further states that although there has been some speculation about how independent students will be in an online learning format, many K–12 students, particularly those with low socioeconomic status, need additional parental support to operate successfully in online learning. Many barriers prevent families from being involved in their child's education, including financial status, lack of access to technology, and/or a cultural disconnect between parents and schools. According to Chen, schools should provide resources and training to help close the gap, to ensure parents have the tools needed to assist their children's educational pursuits.

Our EWS model was shaped by this idea to encourage using the indirect indicators of parent and student engagement (i.e., poverty level of students and rates of student absences) because family engagement data is unavailable in all school-level datasets. Although these variables are not a measure of engagement directly, there is evidence from previous studies that these variables can be useful as a proxy for engagement if observational data regarding family involvement cannot be obtained.

3.2 Dropout Prevention Practice Guide

The What Works Clearinghouse (WWC, 2017) has established evidence-based practices in support of EWS design objectives. Strategies identified by the clearinghouse include developing student-advocate relationships, forming small learning communities, incorporating academically focused extracurricular activities, and using data-driven methods to identify areas for improvement. All of these strategies focus on creating an individualized and supportive learning community to reduce student dropout risk. WWC also emphasizes ongoing monitoring and data collection to inform the implementation of interventions. Our research incorporates this recommendation by assessing classification effectiveness using data-informed metrics (precision, recall, PR-AUC) to better understand the identification of at-risk schools through the

EWS compared to the sole measure of accuracy.

3.3 Geospatial Analysis and Educational Equity

Cobb (2020) defined geospatial analysis as an analytical strategy to understand the geographical disparities of educational opportunities and access through the use of geographic information systems (GIS) methodology. Through the review of 42 GIS-based studies conducted from 1995-2019, Cobb explains how geospatial data can be visually represented to display inequities in school quality, resource allocations, and student outcome measures, which may be invisible or obscured in aggregate reports. Cobb concluded that while geospatial analysis can identify where these inequities occur, it can also serve as a basis for evidence-based policy making through the interconnection of social / economic / geographic dimensions of educational equity.

We will build on Cobb's (2020) findings by using a spatial dimension in our analysis and mapping predicted risk at the school level throughout California. This spatial dimension of our analysis not only provides context for the model output, but it will also allow policymakers to visually represent which areas of the state have the lowest high school graduation rates and, therefore, enable targeted interventions and fair distribution of educational resources.

4 Methodology

To develop a statewide, school-level EWS to predict high school graduation outcomes, we needed to create a unified dataset built from publicly available indicators aligned with the ABC framework. This step was necessary because California's education data is not stored in a single repository. It is spread across multiple state reporting systems that differ in structure, file format, reporting categories, and reporting years. Several datasets, such as the graduation, absenteeism, staffing, and socioeconomic files, were available for the 2021–22 academic year. However, the school

safety and climate data were only publicly accessible for earlier years. Therefore, all datasets required filtering and temporal alignment before they could be merged into a single analytic file for modeling.

4.1 Data Acquisition and Aggregation

Data acquisition and merging were performed using Python (pandas) within a Jupyter Notebook environment. Publicly accessible datasets were acquired from the CDE for the 2021-22 academic school year. Each dataset was downloaded in its native format, such as CSV, XLSX, or TXT files, and imported into Python for initial cleaning and merging. We used the 14-digit County-District-School (CDS) code, which serves as the unique identifier for joining each dataset into the main combined dataset. Each dataset was filtered for high schools only. Rows that reported on subgroup or aggregate data were removed from the dataset. Schools that did not report at least 90% of their expected cohort enrollment (`pc_hs_enrollment < 90%`) were removed to ensure graduation rate validity. Additionally, any rows missing the target variable (`graduation_rate`) were excluded, since these records could not be used in supervised modeling.

To build the unified dataset, we combined a number of publicly available sources from the CDE. These included:

- Adjusted Cohort Graduation Rate (ACGR)
- Chronic Absenteeism
- FRPM (Free/Reduced-Price Meals)
- CBEDS Staff Assignment
- School Directory / School Characteristics
- CalSCHLS School Safety and Climate (2017 - 2019)

These datasets collectively provide school-level measures aligned with the ABC framework.

In addition to the core 2021–22 datasets, we also incorporated the School Safety and Climate (CalSCHLS) dataset. The most recent publicly available CalSCHLS results were county-level aggregate findings from the 2017–2019 school years. School safety indicators were not collected in the 2019–21 cycle due to pandemic-related school building closures. This makes the 2017–2019 results the most recent data available (Austin et al., 2023). Even though these indicators do not align temporally with the 2021–22 data, prior analyses from the California Healthy Kids Survey (CHKS) show that school climate and engagement measures typically follow multiyear trends rather than showing sudden year-to-year changes. Because these county-level climate indicators provide meaningful contextual information related to the behavioral dimension of the ABC framework, we included the 2017–2019 CalSCHLS data in our combined dataset. These variables were merged at the school level using the county name. The difference in reporting years and the use of county-level rather than school-level climate data are noted and considered during interpretation.

Missing values were retained at this stage and addressed later in the data quality and feature engineering steps. No scaling, normalization, or feature standardization was performed during data acquisition. The datasets were merged into a single table and preserved in raw format. This merged dataset was then saved as a versioned file to ensure reproducibility in subsequent exploratory data analysis (EDA) and modeling phases.

4.1.1 Exploratory Data Analysis

Before modeling could be performed, we conducted EDA to get an understanding of the dataset we were working with. The merged raw dataset used for EDA contained 1,067

school-level observations and 46 columns, including the target variable. This is essential for ensuring that the merged features are aligned with the ABC framework, detecting formatting inconsistencies across the CDE datasets, and identifying potential issues such as missingness, skewness, outliers, or FERPA-related suppression that could affect model performance.

We performed both quantitative and visual EDA using Python (pandas, seaborn, matplotlib) and our custom 'jcds' library, which provided helper functions for summary statistics and profiling. For each variable, we examined summary statistics, plotted distributions, and analyzed missingness patterns. Because the dataset included both percentage-based measures, count data, and categorical identifiers, we reviewed each variable type separately. Visualizations included histograms, boxplots, countplots, and correlation heatmaps.

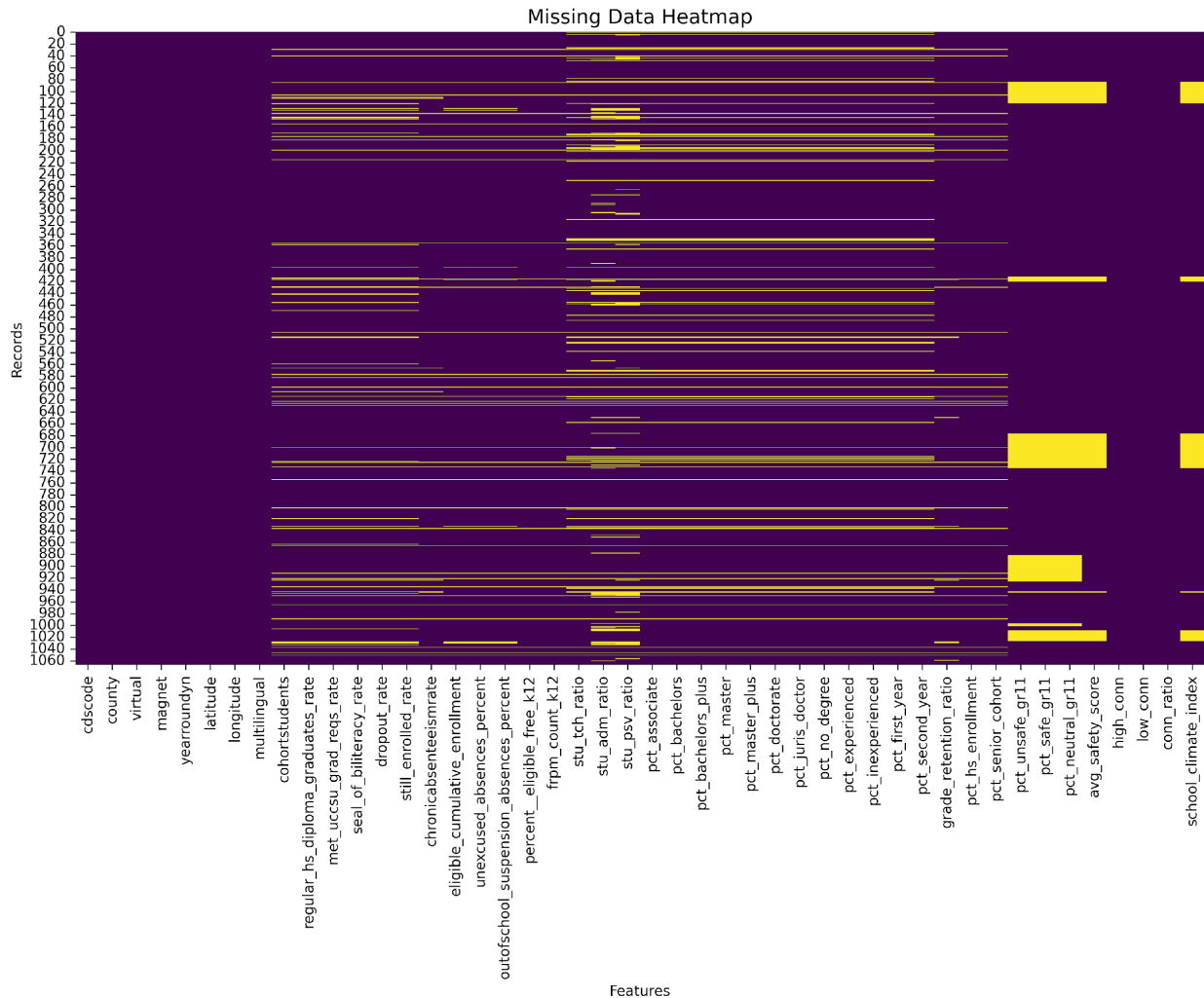
General EDA confirmed that the 2021-22 datasets aligned structurally after cleaning and that most variables followed interpretable distributions consistent with known statewide education trends. For the CalSCHLS school climate indicators, we identified two different types of measures: *Perception of School Safety by Grade Level* and *School Connected* indicators. These observations informed later decisions about which variables were appropriate to retain during data preparation.

4.1.1.1 Initial Observations on Missingness

As part of the general EDA, we reviewed all variables with more than 10% missing values to determine whether the gaps reflected random omissions or systemic reporting issues (see Figure 1). These variables came from three sources: student-staff ratio, safety perception by grade level, and safety perception by school connectedness.

Figure 1

Missingness Heatmap



Further inspection revealed that the missingness patterns were attributable to data source limitations rather than random data loss. The student-staff ratio field is inconsistently reported in the CBEDS staff assignment dataset, particularly among smaller or newly established schools. The School Safety and Climate indicators, derived from county-level CalSCHLS aggregates, contained structural gaps because some counties did not report all perception measures for all grade bands, especially during the 2017-2019 school years.

Because these missing values primarily reflected systematic reporting practices rather than random omissions, they were retained at this stage and revisited in the data quality and feature engineering steps, where we

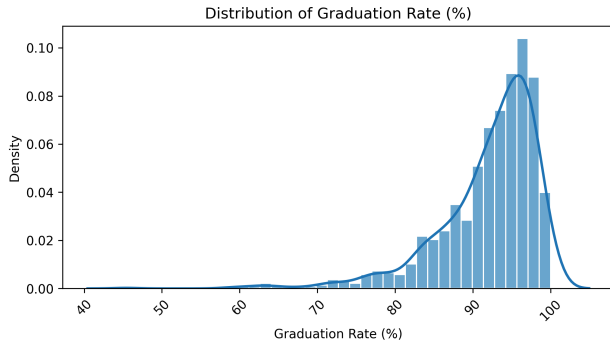
determined how to handle each variable during modeling.

4.1.1.2 Univariate Analysis

Univariate analysis was used to review the distribution and variability of each feature. The primary target variable, `regular_hs_diploma_graduates_rate`, was left-skewed, showing that most schools report high graduation rates and, therefore, confirming a strong class imbalance (see Figure 2). Several predictors were also skewed, suggesting that transformation would be needed during modeling.

Figure 2

Density Plot of Graduation Rates (%)

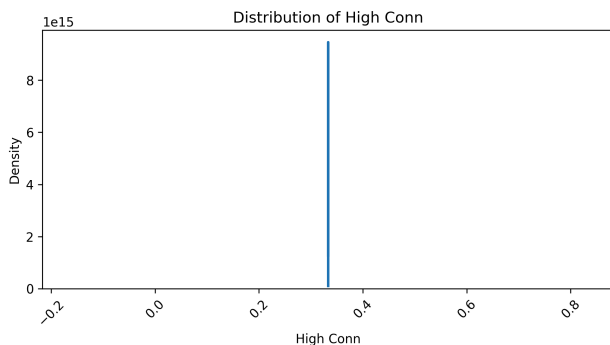


Three school-climate variables showed no variability across schools and were removed: `high_conn` (see Figure 3), `low_conn`, and `conn_ratio`. Geographic identifiers (latitude, longitude, and county) were kept only for mapping and excluded from modeling.

Outlier review showed that `pct_associate`, `pct_no_degree`, and `pct_neutral_gr11` had more than 10% outliers. This was due to heavy right-skewed distributions instead of data errors. Categorical variables such as virtual instruction type, magnet status, multilingual designation, and year-round operation were also reviewed. Most schools fell into a single dominant category, and missing values appeared to reflect reporting gaps rather than random missingness.

Figure 3

Distribution of High Connectivity



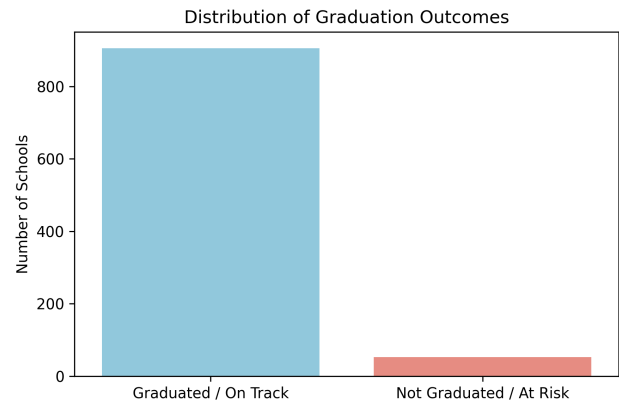
The target variable showed a substantial class imbalance (see Figure 4), with 94.5% of

Figure 5

schools classified as "Graduation / On Track". To address this imbalance, we used stratified train-test splitting and class-weighted models during training without having to use synthetic resampling.

Figure 4

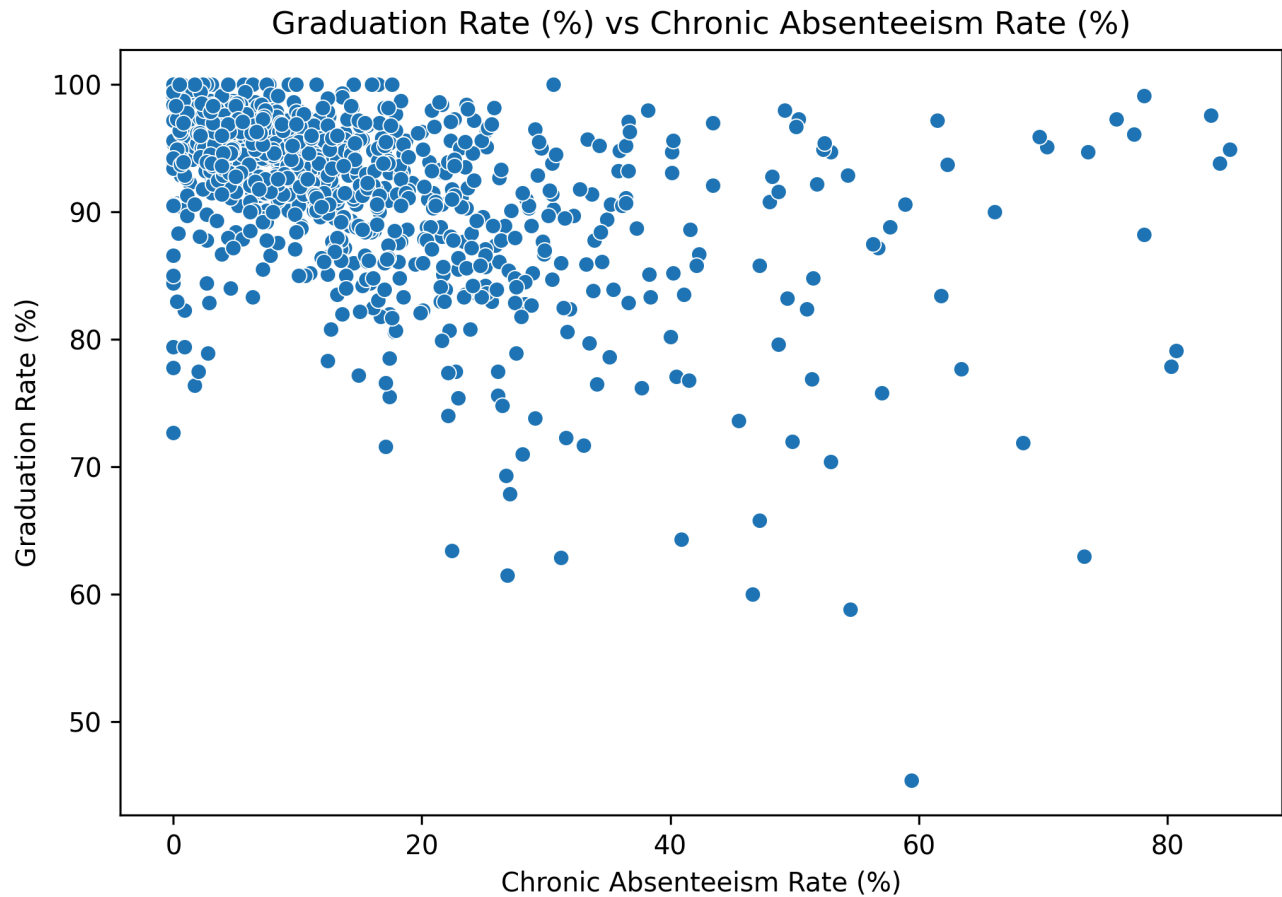
Distribution of Graduation Outcomes



4.1.1.3 Bivariate Analysis

After the univariate analysis, there was a bivariate analysis conducted to see the relationship between key variables and the outcome of graduations. A significant observation was that the strongest and most consistent patterns emerged from the variables that are common with the ABC framework. Variables that reflect attendance showed some of the strongest negative correlations with graduation rate. As shown in Figure 5, there was a consistent pattern between chronic absenteeism and lower graduation rate. There was also a similar relationship where unexcused and suspension-related absences negatively correlated with graduation outcomes at a lower rate. The schools that were higher in FRPM eligibility had a significantly lower graduation outcome, as illustrated in Figure 6.

Scatterplot: Chronic Absenteeism vs Graduation Rate



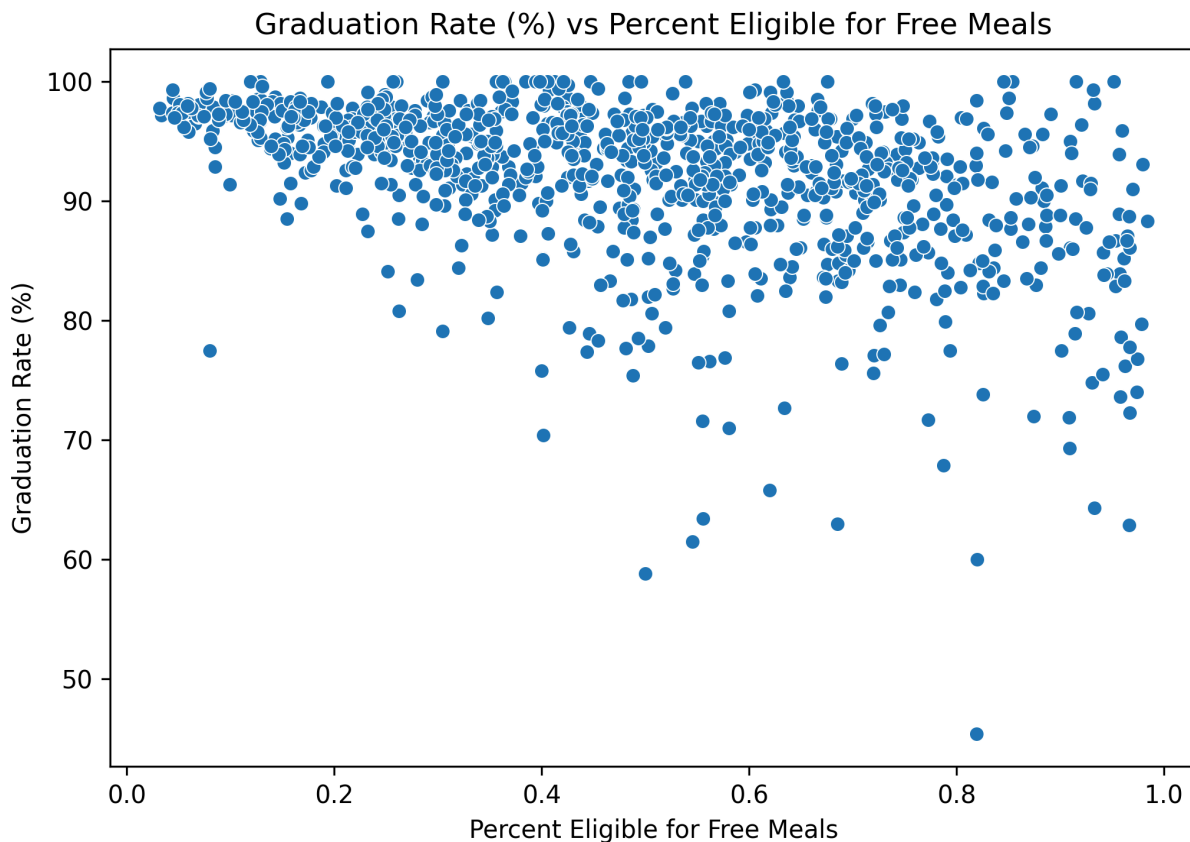
There was insightful information seen from instructional indicators. Schools that had higher percentages of experienced teachers tended to have higher graduation rates. However, schools that had first-year teachers or inexperienced teachers had lower outcomes. There was also a slightly modest positive relationship between student-teacher ratio and graduation rates. In this dataset, it was revealed that schools with large class sizes tended to have slightly higher graduation outcomes. This pattern may reflect a broader contextual factor, such as school size, district resources, or community demographics.

The feature-target correlation supported the visual patterns displayed. Dropout rate and still enrolled rate had the strongest negative correlation with graduation ($r = -0.91$, -0.67

respectively), this is because they directly represent students who did not graduate. To continue, the strongest negative predictors were unexcused absences ($r = -0.47$), chronic absenteeism ($r = -0.43$), and FRPM eligibility ($r = -0.44$). As for the strong positively correlated variables, they were not as high in number but are still notable: A –G completion rate ($r = 0.37$), student-teacher ratio ($r = 0.25$), and the percentage of experienced teachers ($r = 0.18$) were the ones with the largest positive correlation. Safety and climate indicators resulted in small positive correlations which suggest that these variables have an effect on graduation indirectly or through a broad school-level condition as opposed to a direct relationship.

Figure 6

Scatterplot: FRPM vs Graduation Rate

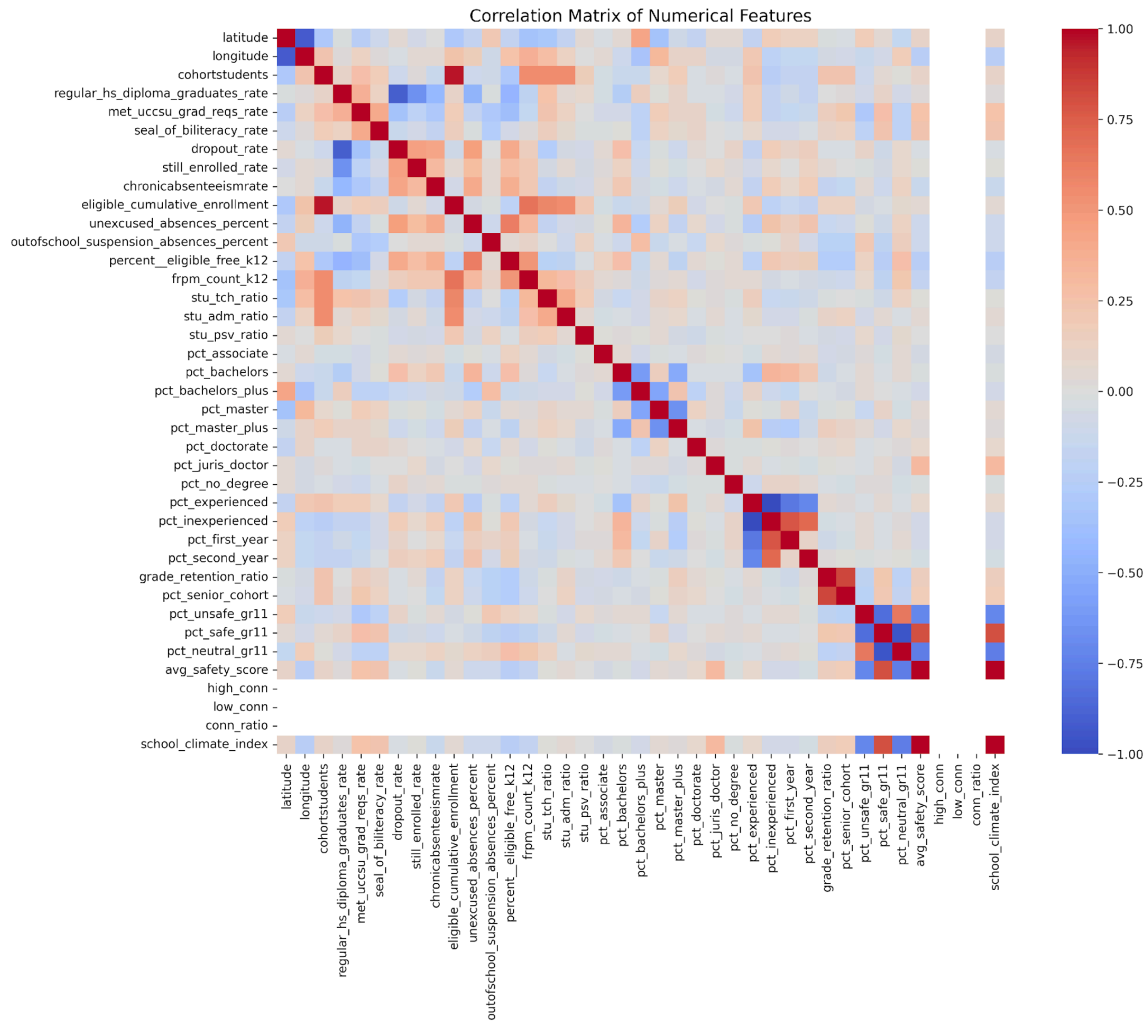


There was a correlation heatmap created to examine the relationship between numerical features, as shown in Figure 7. The highest correlation came from `avg_safety_score` and `school_climate_index` ($r = 1.00$), which demonstrated that both of these indicators are similar. There was a negative perfect correlation between `pct_inexperienced` and `pct_experience` ($r = -1.00$), which is teacher experience and reflects how CDE reports staff categories as mutually exclusive proportions. Features that represented enrollment measures such as `cohortstudents` and `eligible_cumulative_enrollment` also

demonstrated a strong positive correlation ($r = 0.96$). Graduation and dropout rate had a strong negative correlation ($r = -0.91$), confirming internal consistency in the dataset. Safety-related indicators, including `pct_safe_gr11`, were strongly positively correlated with climate scores ($r = 0.81$), while `pct_unsafe_gr11` was strongly negatively correlated with `pct_safe_gr11` ($r = -0.83$). After understanding the heat map and correlations it helped with identifying redundancies, clarifying overlapping constructs and guiding the selection of variables for modeling.

Figure 7

Correlation Heatmap



4.2 Data Quality

To have reliable school-level predictions, a critical step was generating high data quality. Based on the insights gained from the EDA, the next step was to assess the quality of the dataset to understand the strengths, limitations, and readiness for the modeling phase. After cleaning and merging the CDE datasets, it was already generally high in quality and suitable for modeling. There was consistent formatting across the variables, standardized percentage fields, and aligned CDS codes, with aggregate and subgroup rows removed. The features were mostly within the expected numerical ranges, and the key ones related to absenteeism, poverty, school climate, and staffing all had and aligned with expected statewide patterns.

The dataset was mostly complete, however there were some missingness in CalSCHLS climate and safety measures and in some categories of absenteeism, which was a reflection of the limitations in reporting. The missing climate and safety data was handled by creating two different modeling datasets. The first one was a dataset that excluded the climate index variable ($n = 958$ schools), the second dataset excluded seven counties that had data that was unreported ($n = 806$ schools). The other missing values for staff ratios, teacher characteristics, and grade retention, were imputed using the median. This affected 14-59 values per variable. Doing this allowed a maximum sample size to be used while maintaining the integrity of the data for analysis. There were also a small number of outliers,

such as unusually high student-teacher ratios or absenteeism that had zero reported values for the whole school. A number of features also measured similar values, these included overlapping school climate indicators and teacher experience categories. Overall, the dataset provided a reliable foundation for modeling, with remaining limitations clearly identified and manageable during late preprocessing steps. Those steps included: addressing outliers by removing them or capping extreme variables, removing redundant and highly correlated variables using feature selection, and standardizing numerical features to make sure the scales were comparable for model training.

Additionally, all publicly accessible CDE datasets are presuppressed for privacy under FERPA regulations and do not include any personally identifiable information. Small subgroups, such as graduation outcomes, absenteeism categories, discipline incidents, or demographic breakouts with very few students, are intentionally masked by the state to prevent re-identification. This suppression creates some of the structural missingness observed during EDA, particularly in the absenteeism subgroups. In contrast, the limited availability of certain CalSCHLS climate indicators reflects systemic differences in county reporting rather than privacy masking due to FERPA. Because our analysis uses only school- and county-level aggregates, the risk of re-identification is eliminated, and no additional suppression was needed during data preparation.

4.3 Feature Engineering

Feature engineering was performed to prepare a cleaned dataset for modeling and to ensure alignment with the ABC framework. There were several variables that were identified during EDA as non-informative and were removed prior to modeling since they did not contribute meaningful variance or provided no predictive value:

- Derived connectedness variables: high_conn, low_conn, and conn_ratio. These variables showed low variance.
- All categorical features were excluded because EDA showed that each feature showed minimal variation and limited predictive contribution.
- Identification-only variables: latitude, longitude, county, CDS Code. These variables were only used for initial identification and merging the datasets.

These features were removed to ensure generalizability.

Since a number of behavioral indicators came from the CalSCHLS climate surveys, the raw data required additional engineering to convert student perception measures into numeric features that could be incorporated into the model. The original safety indicators were reported as distributions across a five-point Likert scale: “*Very Safe*”, “*Safe*”, “*Neither Safe nor Unsafe*”, “*Unsafe*”, and “*Very Unsafe*.” These ordinal categories were converted into numeric values and combined into an expected-value safety_score, weighing each category on a 1 - 5 scale according to its favorability. This created a continuous measure of perceived school safety at the county level. Another engineered feature, avg_safety_score, was created by averaging safety scores across available grades, providing a stable and interpretable climate feature aligned with the behavior component of the ABC framework. Connectedness-based climate variables were evaluated but ultimately removed due to a lack of variance and limited predictive value.

Several numeric predictors exhibited strong right skews, including absenteeism rates, staffing-related percentages, and climate measures. These variables were left untransformed in the final dataset but were standardized later within the modeling pipeline to prevent data leakage and ensure consistent scaling during training.

The target variable was engineered as a binary label using a 90% graduation rate threshold to serve as an early warning system to flag

emerging risk rather than just underperformance. Schools with graduation rates below 90% were labeled “At Risk” and all others were labeled “On Track.” Since only 26.3% of schools fell into the “At Risk” category, the class imbalance identified during EDA was addressed during modeling. Using a 90% threshold also resulted in a more workable class distribution than the original 80% threshold used during EDA. This allowed for stratified sampling and class-weighting to address the imbalance without the need for using synthetic resampling techniques.

Overall, feature engineering focused on removing non-predictive fields, preparing county-level climate variables for use as behavioral indicators, handling skewed numeric predictors through pipeline-based scaling, and constructing a clear and reproducible binary target variable for classification.

4.4 Modeling

The modeling phase focused on developing a reliable EWS with county-level data. We examined which supervised classification algorithm could best predict a low graduation outcome in the presence of the limitations of real data. While the original data included safety-climate indicators, preliminary testing demonstrated that these variables were missing entirely for seven counties. The missing data was not random and would have required removing those seven counties from the dataset; the safety-climate variables were excluded from modeling.

With the remaining data, random forest was used to determine the strongest predictors of graduation outcomes, as the Random Forest method is robust to nonlinear data, can handle mixed variable types, and yields interpretable feature rankings. Also, to reduce dimensionality, 15 predictors were selected while still preserving predictive performance.

4.4.1 Selection of modeling techniques.

The choice of techniques for developing a model was based on selecting classification techniques that would predict low graduation rates utilizing only county-level statistics. The need to select algorithms that can handle both categorical and continuous data, as well as nonlinear relationships due to the mixed-type variable in the dataset, requires an evaluation of those types of models. The initial application of a Random Forest model served to provide an understanding of feature importance so that the reduction would be guided by the top 15 most important predictors. This involved the application of the following classifier models: logistic regression, random forest, naïve bayes, support vector machine (SVM), XGBoost, K-nearest neighbors (KNN), and decision tree; as these are representative of a wide variety of classification methods, including linear, nonlinear, probabilistic models etc. Using a large number of different models provided a more accurate determination of which models are best suited for the early warning predictive task of identifying counties that fall as the risk below the 90% graduation threshold.

4.4.2 Test design, i.e. training and validation datasets.

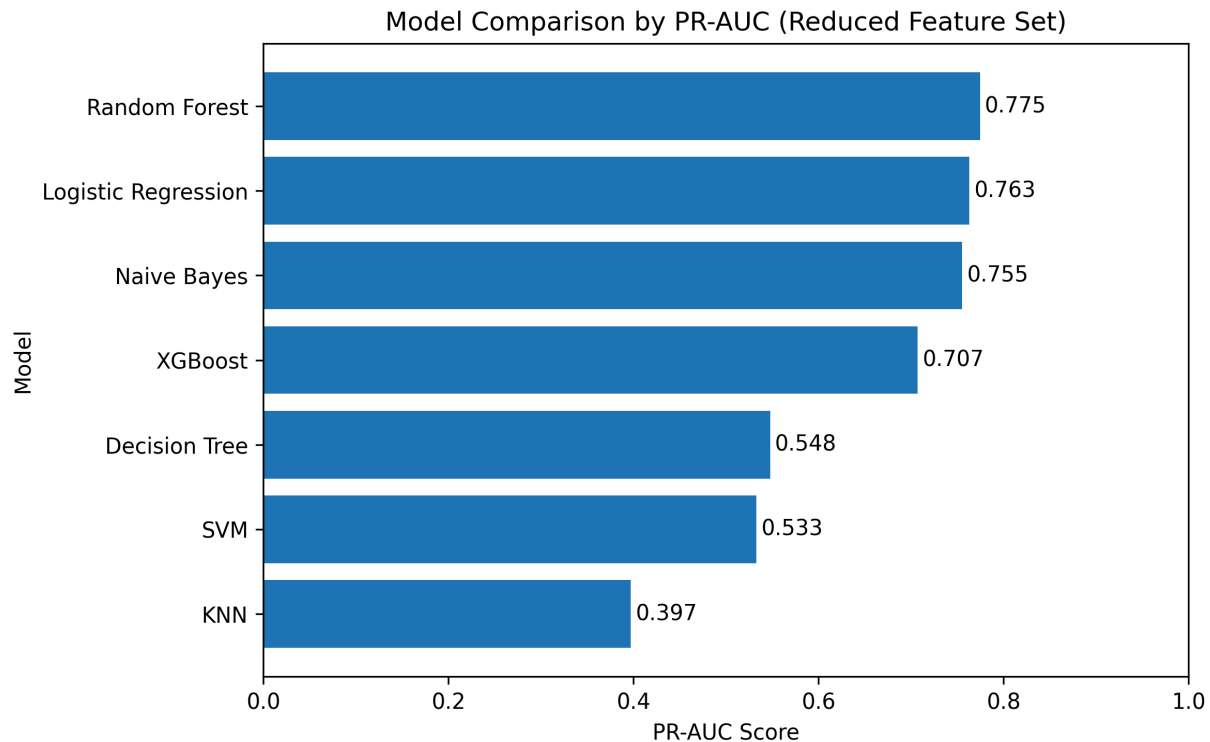
Each model was trained with 80/20 train-test splits. The training set was used to fit the classifiers, while the remaining 20% was held out as unseen data for evaluating performance. As the target variable was imbalanced, accuracy metrics such as AUC weren’t meaningful for evaluation. Thus, rather than prioritizing accuracy, precision recall scores, F1 score, and PR-AUC were chosen so that we could better understand how performant the models would be on predicting many of the counties. So, no oversampling or synthetic balancing methods were applied to the data. All classifiers were run using their default hyperparameters. Our goal was to compare baseline model performance rather than optimize the individual algorithms.

With the final chosen model, we also developed a Streamlit web application that interactively displays graduation-risk predictions based on

user-selected school indicators. The app is available at: <https://ca-early-warning-system.streamlit.app/>.

Figure 8

Model Comparison by Precision-Recall AUC Performance



5 Results and Findings

The modeling results show that county-level data can predict low graduation outcomes when reduced to the most informative predictors. After removing the safety-climate variables with the seven missing counties, the data had 25 usable features. The top 15 predictors identified by using a Random Forest were used. These top predictors captured academic, attendance, and socioeconomic patterns, such as `still_enrolled_rate`, `chronicabsenteeismrate`, `uexcused_absences_percent`, `met_uccsu_grad_reqs_rate`, and `percent_eligible_free_k12`, etc.

Figure 8 illustrates the precision-recall AUC performance of all evaluated models using the reduced feature set. Since interpretability was

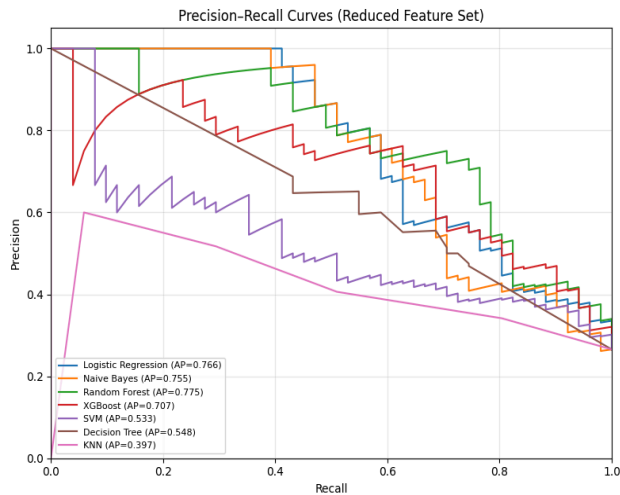
not the highest priority and the predictions will not directly impact individual students, it is reasonable to select the model that maximizes predictive power. The models that performed the best and showed stable precision-recall performance on imbalance outcomes included Random Forest, Logistic Regression, and Naïve Bayes- they all achieved PR-AUC scores of 0.75-0.79. All three models demonstrated better performance than Decision Trees and KNN. The Support Vector Machine model did achieve a perfect precision of 1.00; however, the model's recall was very poor at 0.059, meaning that the SVM identified nearly no at-risk schools.

Overall, the random forest model is the most robust and reliable of the models because it achieved the highest PR-AUC and strong precision-recall performance, it also maintained

consistent performance across the entire curve, and it outperformed other models on the imbalance classification metrics (see Figure 9). In addition, the logistic regression model also provides a simple alternative.

Figure 9

Precision-Recall Curves for All Models (Reduced Feature Set)



5.1 Evaluation of Results

All evaluations were conducted using measures relevant to the imbalance data. These include

Table 1

Performance Metrics for All Classification Models

Model	Precision	Recall	F1 Score	PR-AUC
Random Forest	0.720	0.706	0.713	0.775
Logistic Regression	0.542	0.765	0.634	0.766
Naïve Bayes	0.547	0.686	0.609	0.755
XGBoost	0.702	0.647	0.673	0.707
Decision Tree	0.500	0.725	0.592	0.548
SVM	1.000	0.059	0.111	0.533

Precision, Recall, F1 score, and PR-AUC (see Table 1). In terms of Precision and Recall, and the highest PR-AUC, Random Forest was the best of all evaluated models for determining which counties are at greater risk of low graduation rates. Additionally, Logistic Regression performed well, making it a practical secondary option due to its easy interpretation for decision makers.

The model's top predictor variables, including still-enrollment rates, chronic absenteeism, FRPM eligibility, and failure to complete graduation requirements, also reflect the risk factors for dropping out of school in literature examining student socioeconomic status, engagement, and academic pathways (Chen et al., 2019; Sava et al., 2017; Siegle et al., 2016). Therefore, while the model performed statistically, it is likely that it has captured additional dropout risk patterns that were previously understood through other research. These findings support our original hypothesis that schools with higher absenteeism and greater socioeconomic disadvantage are more likely to have lower graduation rates.

KNN

0.517

0.294

0.375

0.397

Note. PR-AUC = precision–recall area under the curve. Higher values indicate stronger performance on imbalanced outcomes.

6 Discussion

This study showed that data that is publicly available while being at the school level can be used to effectively predict graduation outcomes and look at a crucial gap in educational early warning systems. The EWS models that already exist mostly rely on data that is protected and at the student level. With this study, it was revealed that aggregated indicators are still capable of having a strong predictive power when it is analyzed by the ABC framework, as well as with a fourth school level category that emerged: Support and Services. This category is information about how much experience a teacher has, the available staff, and how much resources a school has to support its students, which all contribute to the educational conditions students are in.

Random forest, logistic regression, and naïve bayes were the best-performing models and achieved a PR-AUC score of 0.75-0.79. This demonstrates how they have strong predictive capabilities for identifying schools at risk of low graduation rates, and that need the extra support. The predictors that were the strongest were similar to those in well-established dropout literature. The strongest negative correlation with graduation was attendance-related variables, more specifically, it was chronic absenteeism and unexcused absences. This confirms that absenteeism is an early warning sign and is one of the determinants that require quick action. Another significant predictor was FRPM eligibility, which captures poverty. This confirms that students who are disadvantaged in their socioeconomic status have different opportunities and resources, which affect their educational performance.

Another indicator that is important to look at is course performance. Students' rate of meeting UC/CSU A-G requirements had a significant impact on determining graduation status. There

was an influence on whether a student graduated on time based on the rigorous coursework and the access to academic pathways. Another critical variable was `still_enrolled_rate`, which provides information on whether a student has continued to be enrolled past the typical four years without graduating. This variable gives insight into a possible gap in the academic system or support. The additional insight found in this study was that school-level support and services were a contributing factor. The correlation between teacher experience and student graduation was moderate; however, it was included in the top predictors, and it suggests that teacher preparedness and instructional stability are part of a broader system that influences how students perform.

The results from this analysis also strongly connect to existing literature. The ABC framework still continued to show up at the school level in this analysis, which shows that based on the results from using student level data are part of a broader organizational pattern. Chen (2019) discussed how important it is for parents to be involved in student academics and based on the FRPM and absenteeism rates there was an indirect parallel. These variables can give insight into family-level challenges and their socioeconomic status which also results in some information on how involved a parent can be depending on their situation. The recommendations from the What Works Clearinghouse (2017) about looking at the bigger picture was followed in this study. It was suggested to use different indicators to identify risk, therefore a combination of attendance, behavior and course related factors were used. The schools in our dataset mostly had students who graduated and that caused an imbalance in the outcome. With this in mind, the precision and recall were looked at to help identify the model that was more capable of identifying the risk.

6.1 Limitations

There were a few limitations that were encountered in this study. One limitation was not being able to use the climate data as it was outdated (2017–2019), reported only at the county level, and it was missing for seven counties. The school climate indicators provide information on the school environment a student is in, this has to do with their feelings of safety, connectedness, and inclusion. These are all important factors that can measure engagement, but due to the limitation it was not useful, and it did not improve the performance of the model. This resulted in dropping the climate variables from the final model, which still performed well and had a more reliable set of predictors. Another limitation was the lack of variation that school level data gave, it did not provide information on more specific grouping of students. Lastly, the data used was all from California, which does not have the same reporting system as other states. There would need to be adjustments made for those states based on their reporting structure and the data that is available.

6.2 Conclusion

This study developed a school-level EWS that is capable of predicting low graduation outcomes using only publicly available data that is at the school level from the CDE. The strongest predictors that came from this study were: chronic absenteeism, unexcused absences, FRPM eligibility, still-enrolled rates, and A-G completion. The variables are ones that are common in research that describe how attendance, socioeconomic obstacles, and preparation in academics have an association with graduation outcomes. There was another variable that emerged which was Support and Services. This gave insight into how teacher experience had a moderate effect on graduation rates in a broader sense.

The performance of these models had PR-AUC scores between 0.75-0.79. This signifies that a

reliable early warning predictor can be done by using public school-level data and does not have to be student-data that is private. Due to this, the approach allows for more accessibility from districts of all sizes while also being in compliance with FERPA. There are numerous studies revolving around student level data that this study adds to it at a school level and is helpful in identifying the schools that are experiencing challenges with regards to the system. Finally, the findings from this analysis revealed that using data that is open and not sensitive can be useful in decision making that is more equitable. It will also provide help with early interventions for all of California's diverse schools and give insight into where to allocate resources.

Overall, this study provided insight on the usefulness of school level predictions which are more attainable and practical. With the publicly available data, we were able to combine relevant indicators. The Early Warning System gives districts and schools a tool they can use to confirm when there is support needed and be able to intervene early enough before the graduation rates decrease.

6.3 Recommend Next Steps / Future Studies

This EWS has different opportunities in which it can be worked on for future studies. First, gathering and using climate indicators for the schools that are up to date can improve future models. This represents how safe a school feels for students and their sense of belonging, as well as the support they have with staff and other students. With newer and complete CalSCHLS data, the study will be able to expand to better capture the patterns in behavioral and engagement.

Another recommendation is to have more information on the support and services dimensions, such as number of counselors that are available, what access do schools have in academic intervention, or how available mental health services are for students. Another suggestion is to use data from several years

which can help see the difference in schools over time and the impact specific programs had on graduation rates.

The state of California was able to provide a lot of information for their schools, but many other states have removed or reduced the datasets in education due to privacy policies or other regulations, and it will need a different approach. Using the early warning system in another state will require an adjustment in the model based on the data available.

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